

**What makes an
asset the
dominant
intermediary?**



**Centre for Blockchain
Technologies**

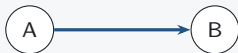
The Making of Dominant Vehicle Currencies

Evidence from DeFi
Jiahua Xu

Transaction-level route structures

Direct swap

one transaction



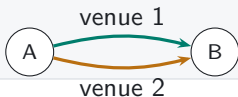
Sequential vehicle route

one transaction



Parallel split

one transaction



Round trip

one transaction



Swap events in one transaction reveal sequential routes, parallel splits, and round trips.

Measuring vehicle dominance

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k .
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.



Count-weighted dominance

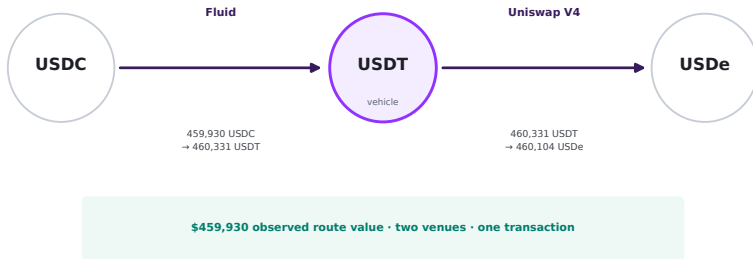
Share of indirect routes that pass through k

Value-weighted dominance

Share of comparable routed value that passes through k

One transaction reveals cross-venue vehicle use

OBSERVED ROUTE



USDT is observed in the middle of this route. Whether the chosen route beat a direct alternative requires the exact pre-trade opportunity set and is not inferred from this transaction trace.

Architecture changes routing opportunities

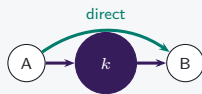
V1 mandatory hub



Only ETH-token pools: token-to-token execution must pass through ETH.

Hub use is imposed

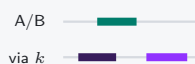
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

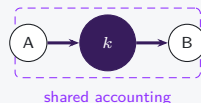
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



Two pools still execute; intermediate token movement may be netted.

Settlement and vehicle use can diverge

Architecture feasible rules

Exposure realised pool and route use

Calendar date timing only

The panel follows the market from V1 through V4

DATA



Routes are reconstructed transaction by transaction across the routed venue perimeter. Daily and weekly panels are derived from the same route units.

Four measures capture use, scale, and network position

1. Vehicle share

Fraction of route units intermediated by asset k .

2. Excess use

Intermediary share relative to endpoint demand.

3. Network position

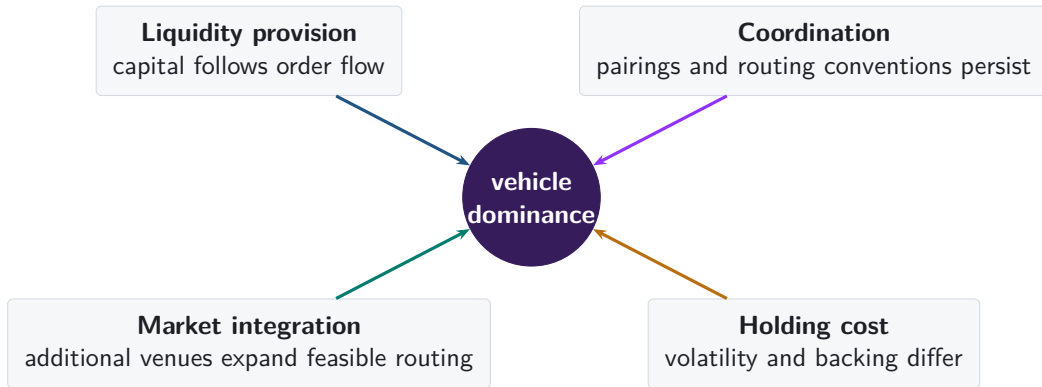
Degree, strength, and betweenness on the trading graph.

4. Execution advantage

Direct-route output relative to a feasible vehicle route at the same pre-trade state.

Together, the measures distinguish use, scale, network position, and execution cost.

Four forces can sustain vehicle dominance



The empirical design separates formation, rotation, and persistence

FORMATION

Architecture changes

Imposed intermediation, arbitrary pairing, concentrated liquidity, and singleton settlement.

ROTATION

Panel variation

Count share, value share, excess use, network position, and route complexity over time.

PERSISTENCE

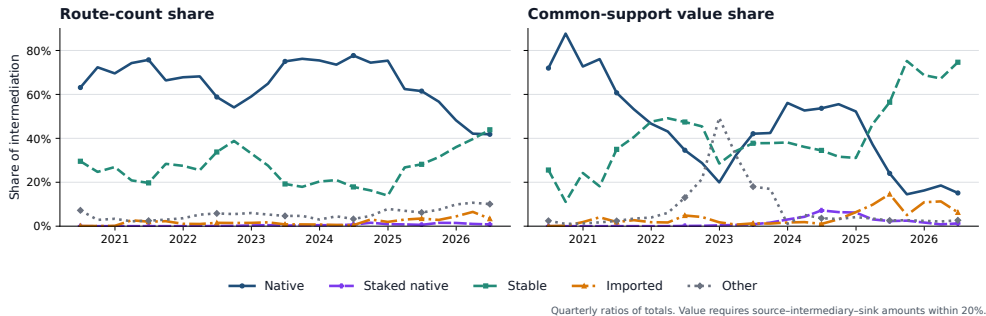
Contemporaneous alternatives

An incumbent retains volume despite a cheaper feasible route.



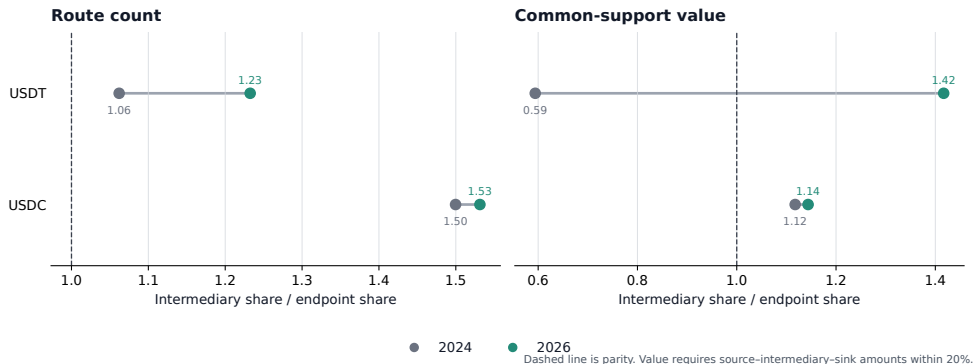
The vehicle mix rotates; it does not trend monotonically

The intermediating asset changes over time



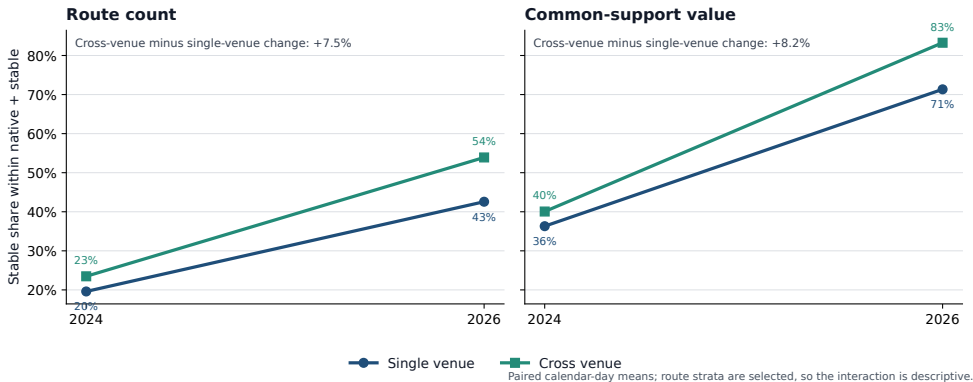
Calendar quarter describes when the mix changes; it does not identify which contemporaneous routing design caused the change.

USDT is the 2024–2026 transition margin



USDC begins above parity and moves little. USDT crosses parity on common-support value; the intermediary-minus-endpoint share gap is the inferential outcome, while the ratio is its relative-intensity translation.

Stable share rises within both single- and cross-venue routes



The stable share rises within both route strata. The larger cross-venue change rejects venue mix as a complete composition account, but selected route strata do not identify a causal effect of integration.

Challenger displacement gradients are strong

30-day outcome	$\hat{\beta}$	SE	t
Challenger vehicle-share response	4.825	0.366	13.17
Incumbent vehicle-share response	-3.872	0.420	-9.22
Challenger overtakes incumbent	9.947	0.486	20.48

$N = 57,989$ pair-candidate observations; 2,213 date clusters. Exact alternating-projection absorption and CR1 rank correction.

What it says: the sparse challenger panel contains strong displacement gradients.

What it does not say: these gradients do not identify transition, routing efficiency, or persistence because the risk set does not yet condition on fixed feasible alternatives.

Architecture exposure lacks isolated substitution events

	5%	10%	25%
Sustained entries	162 (0)	139 (0)	78 (0)
Substitution exits	79 (0)	69 (0)	54 (0)
Role disappearances	7 (3)	7 (3)	7 (3)
Immediate entry contrast	—	—	—
Immediate exit contrast	—	—	—

WHAT THE COMPARISON ESTABLISHES

Zero clean substitution windows. Realised V4 share is endogenous, the pair-week centering is mechanical when shares sum to one, and role appearance/disappearance magnitudes are selected lifecycle endpoints. The design blocks an effect instead of manufacturing one.

The identifying comparison requires raw focal share on an ex ante feasible risk set, with exact-state reversals separating design response from calendar coincidence.

Three questions organize the evidence

1. **Formation**

When does market architecture create or strengthen an intermediary role?

2. **Rotation**

Which changes reflect intermediary demand, endpoint demand, trade size, or routing technology?

3. **Persistence**

How long does incumbent intermediation survive when contemporaneous alternatives are available?

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

A2. Backing regimes vary over time

Native platform asset

ETH, WETH; volatile settlement asset and protocol-native demand.

Fiat-reserve stablecoin

USDC, USDT; reserve and redemption design.

On-chain collateralised

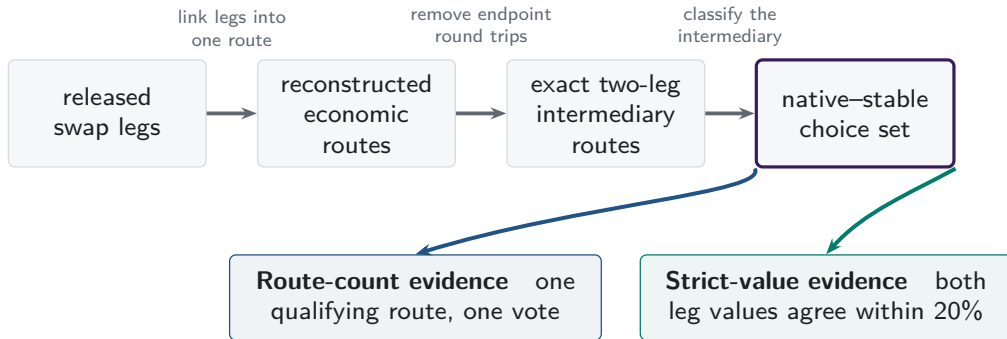
DAI, LUSD, crvUSD; collateral, liquidation, and backing changes.

Tokenised external asset

WBTC, tokenised gold; wrapped claim on an external asset.

Heterogeneity tests use each asset's contemporaneous backing regime.

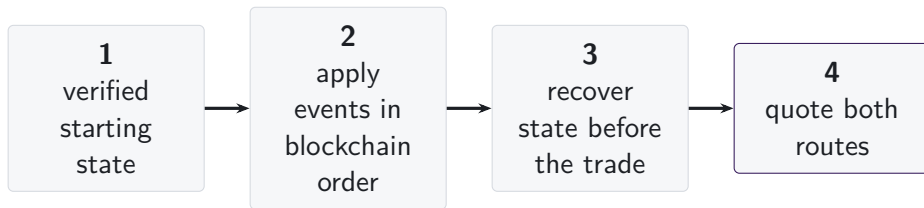
A3. One route universe supports two measurements



Topology-valid routes remain in count evidence when dollar support is unavailable.
Box widths do not represent attrition.

Missing dollar support narrows value-weighted evidence; it does not erase a topology-valid route from the count evidence.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

The common object is the state immediately before execution.
The fields required to recover that state vary with the pricing rule.

■ required state

□ hooks outside comparison

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.



same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions

- | | |
|----------------------------------|---|
| 1. Primary panel | day observations; date fixed effects |
| 2. Calendar sensitivity | day observations; week and weekday fixed effects |
| 3. Aggregation robustness | complete weeks; week fixed effects |
| 4. Transition series | exact 1-, 7-, 30-, and 120-day links; HAC inference |

Weekly ratios are formed from weekly sums; results are reported for weeks ending on each day of the week.

A8. Count and value define separate dominance margins

Count-weighted share

$$S_{k,t}^N = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

Each reconstructed route carries one unit of weight.

Value-weighted share

$$S_{k,t}^V = \frac{\text{Vol}_{k,t}^I}{\sum_j \text{Vol}_{j,t}^I}$$

Each route carries reliably priced dollar value on common support.

Differences between count and value shares may reflect trade size, route structure, or the mix of intermediary assets.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$S_{k,t}^I = \frac{N_{k,t}^I}{\sum_j N_{j,t}^I}$$

How often asset k sits inside a route.

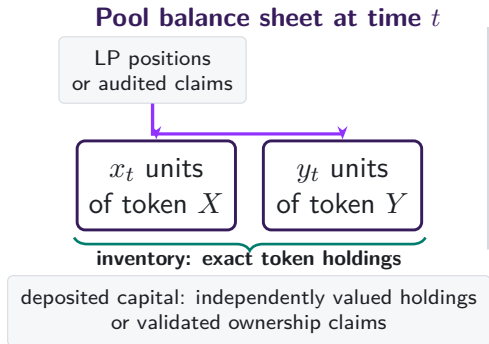
Endpoint share

$$S_{k,t}^E = \frac{N_{k,t}^{\text{in}} + N_{k,t}^{\text{out}}}{\sum_j N_{j,t}^{\text{in}} + N_{j,t}^{\text{out}}}$$

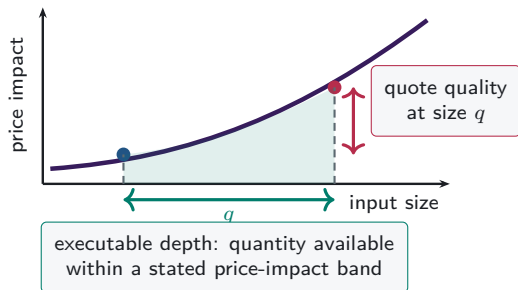
How often asset k is what traders buy or sell.

Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are different objects



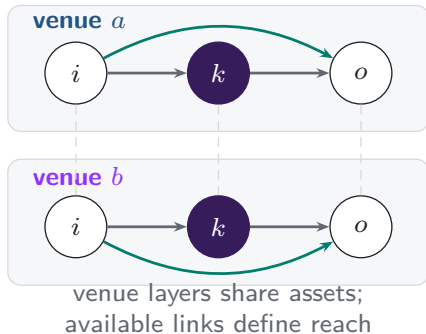
Execution from that state



Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Integration changes the routes that can be chosen

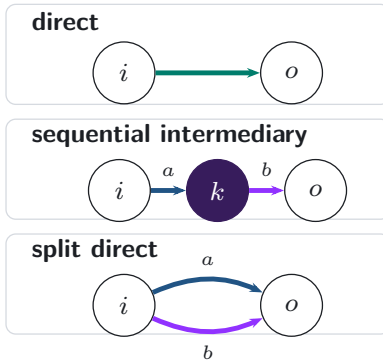
Feasible venue layers



realised
execution



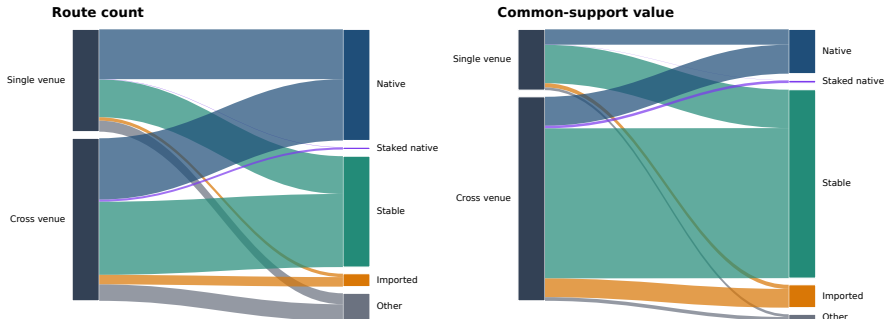
Realised route topology



The analysis therefore separates opportunity-set reach from the composition of realised routes.

A11b. Venue scope and vehicle type differ in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

The ribbons classify realised intermediation jointly by venue scope and intermediary type. They describe selected routes; they do not identify an effect of integration or the unchosen opportunity set.

A12. References: vehicle currencies and market structure

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A13. References: decentralised exchange design

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- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
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